

Question 1. Express $\frac{dw}{dt}$ as a function of t .

(1) $w = x^2 + y^2$, $x = \cos t$, $y = \sin t$
 $w = \cos^2 t + \sin^2 t = 1$ $\frac{dw}{dt} = 0$

(2) $w = e^x - \sin y$, $x = t^2$, $y = t^3$
 $\frac{\partial w}{\partial x} = e^x$ $\frac{\partial w}{\partial y} = -\cos y$ $\frac{dx}{dt} = 2t$ $\frac{dy}{dt} = 3t^2$
 $\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}$
 $= 2te^{t^2} - 3t^2 \cos t^3$

(3) $w = \ln((2x + 3y)^2)$, $x = e^t$, $y = t$
 $w = 2 \ln(2x + 3y)$
 $\frac{\partial w}{\partial x} = \frac{2}{2x+3y} \cdot 2$ $\frac{\partial w}{\partial y} = \frac{2}{2x+3y} \cdot 3$ $\frac{dx}{dt} = e^t$ $\frac{dy}{dt} = 1$
 $\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} = \frac{2}{2e^t+3t} (2e^t + 3) = \frac{2}{2e^t+3t} (2e^t+3)$

(4) $w = xy + (xy)^2$, $x = t^2$, $y = \ln t$
 $\frac{\partial w}{\partial x} = y + 2xy^2$ $\frac{\partial w}{\partial y} = x + 2x^2y$ $\frac{dx}{dt} = 2t$ $\frac{dy}{dt} = \frac{1}{t}$
 $\frac{dw}{dt} = (y + 2xy^2)(2t) + (x + 2x^2y)\left(\frac{1}{t}\right)$
 $= 2t \ln t + 4t^3 (\ln t)^2 + t + 2t^3 \ln t$

(5) $w = z + z^2$, $z = xy$, $x = t^2$, $y = \ln t$
 $\frac{dw}{dz} = 1 + 2z$ $\frac{\partial z}{\partial x} = y$ $\frac{\partial z}{\partial y} = x$
 $\frac{dw}{dt} = \frac{dw}{dz} \left(\frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \right)$
 $= (1 + 2z)(y \cdot 2t + x \cdot \frac{1}{t})$
 $= (1 + 2t^2 \ln t)(2t \ln t + t)$ Note: Same as (4)

(6) $w = \frac{z}{x} + \frac{z}{y}$, $x = \cos^2 t$, $y = \sin^2 t$, $z = \ln t$
 $w = z \cdot \left(\frac{x+y}{xy} \right) = \frac{z}{xy}$ because $x+y=1$
 $\frac{\partial w}{\partial x} = -\frac{zy}{(xy)^2}$ $\frac{\partial w}{\partial y} = -\frac{zx}{(xy)^2}$ $\frac{\partial w}{\partial z} = \frac{1}{xy}$
 $\frac{dw}{dt} = -\frac{zy}{(xy)^2} \cdot (-2 \cos t \sin t) - \frac{zx}{(xy)^2} (2 \sin t \cos t) + \frac{1}{xy} \cdot \frac{1}{t}$
 $= 2 \ln t \cdot \left[\frac{1}{\sin t \cos t} - \frac{1}{\sin t \cos t} \right] + \frac{1}{t \cos^2 t \sin^2 t}$

Question 2. A differentiable function $f(x, y)$ has a critical point (maximum, minimum, or saddle point) at (x_0, y_0) if $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$. Determine the critical points for each of the following $f(x, y)$.

(1) $f(x, y) = \frac{x^2}{2} + xy + \frac{y^3}{3}$
 $\frac{\partial f}{\partial x} = x + y$ $\frac{\partial f}{\partial y} = x + y^2$
 We want $x + y = 0$ and $x + y^2 = 0$
 $\Leftrightarrow x = -y$ and $y^2 = y$
 $\Leftrightarrow x = -y$ and $(y=0 \text{ or } y=1)$
 Critical points are $(0, 0)$ and $(-1, 1)$

(3) $f(x, y) = g(u(x, y), v(x, y))$
 where $g(u, v) = u^2 - 4u + v^2$,
 $u(x, y) = x + \ln y$, and $v(x, y) = x - \ln y$.

$\frac{\partial f}{\partial x} = \frac{\partial g}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial x} = (2u-4)(1) + (2v)(1)$
 $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial y} = (2u-4)\left(\frac{1}{y}\right) + (2v)\left(-\frac{1}{y}\right)$

Thus $\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) = 0 \Leftrightarrow (2u-4) + 2v = 0$ and $(2u-4) - 2v = 0$
 $\Leftrightarrow u(x, y) = 2$, $v(x, y) = 0$
 $\Leftrightarrow x + \ln y = 2$
 $\Leftrightarrow x = 1$, $\ln y = 1$
 $\Leftrightarrow x = 1$, $y = e$

The only critical point is $(1, e)$.

(2) $f(x, y) = \left(\arctan(x^2 - 4x + y^2) + \frac{\pi}{2} \right) e^u$
 Let $g(u) = \left(\arctan(u) + \frac{\pi}{2} \right) e^u$ and $u(x, y) = x^2 - 4x + y^2$
 so $f(x, y) = g(u(x, y))$
 $\frac{\partial f}{\partial x} = \frac{dg}{du} \frac{\partial u}{\partial x}$ $\frac{\partial f}{\partial y} = \frac{dg}{du} \frac{\partial u}{\partial y}$ note that $\frac{dg}{du}(u) \neq 0$ for all u , so
 $\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) = 0 \Leftrightarrow \frac{\partial u}{\partial x}(x, y) = \frac{\partial u}{\partial y}(x, y) = 0$
 $\Leftrightarrow 2x - 4 = 2y = 0$
 $\Leftrightarrow x = 2$, $y = 0$
 The only critical point is $(2, 0)$.

(4) $f(x, y) > 0$ implicitly defined by the equation $xf(x, y) + (f(x, y))^2 - x^2 - y^2 = 1$.

Take $\frac{\partial}{\partial x}$:

$f + x \frac{\partial f}{\partial x} + 2f \frac{\partial f}{\partial x} - 2x = 0$

Take $\frac{\partial}{\partial y}$:

$x \frac{\partial f}{\partial y} + 2f \frac{\partial f}{\partial y} - 2y = 0$

Thus $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0 \Leftrightarrow f(x_0, y_0) = 2x_0 > 0$ and $2y_0 = 0$

$\Leftrightarrow 2x_0^2 + 4x_0^2 - y_0^2 = 1$ and $x_0 > 0$ and $y_0 = 0$

$\Leftrightarrow x_0 = \frac{1}{\sqrt{5}}$ and $y_0 = 0$

The only critical point is $\left(\frac{1}{\sqrt{5}}, 0\right)$